

Alan Harris

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<https://Davidah121.github.io>

EDUCATION

Bachelor of Computer Science, Georgia Southern University, Statesboro GA

- August 2018 – December 2020
- Graduated Manga Cum Laude
- Overall GPA: 3.73
- Institutional GPA: 3.87
- Minor in Mathematics

Master of Computer Science, Drexel University, Philadelphia PA

- September 2021 – June 2024
- A Focus in Computer Graphics / Computer Vision / Artificial Intelligence
- Overall GPA: 3.91

EXTRACURRICULAR ACTIVITIES & EMPLOYMENT

- Computer Science Intern, Warner Robins Airforce Base, Warner Robins GA
 - June 2020 – August 2020
 - Worked using C# and WPF to provide GUI development using standard practices.
- Aurora Game Development Club, Georgia Southern University
 - Fall 2018 – Fall 2020
 - Served as project lead in Fall 2018 and as a general developer providing music and programming for projects each semester after.

SKILLS & CERTIFICATIONS

- Game Programming Certificate
 - Awarded by Georgia Southern University in December 2020
- Proficient in C, C++, Java, Game Maker Studio, Visual Studio, Visual Studio Code
- Knowledgeable in C#, OpenGL, Windows SDK, Unity, Git, JavaScript, HTML, CSS, Microsoft Unit Test Framework, Catch2 Unit Test Framework
- Experience with Python, MATLAB, DirectX, PHP, PHP Unit
- Experience with Machine Learning and building AI
- High Performance Computing skills
- Project Management skills

PROJECTS

- SMPL (<https://github.com/Davidah121/SMPL>)
 - A general library created in C++ that helps speed up development with useful classes that are designed to be simple to use but also provide control and performance if desired. It provides some image manipulation functions such as Floyd Steinburg Dithering, Color Quantization, a GUI system, Edge Filters, and other helpful math functions.
 - Primarily serves as a place to experiment with and implement interesting research
- PM 3.0 (<https://github.com/Davidah121/PM3.0>)
 - A web based password manager that is an updated version of two previous iterations that were entirely CLI based and clunky to use across multiple devices.
 - It is intended to not only be usable in a real world scenario, but also something you locally host and maintain with a simple enough API that you could develop your own interface to use it instead of the default interface provided.
 - Can use encryption and HTTPS but allows the programmer to implement their own desired encryption through a simple wrapper. Personally, I use my own encryption method
- Building an Efficient HashTable (<https://github.com/Davidah121/Building-An-Efficient-HashTable>)
 - An overview of the process of building a highly efficient HashSet or HashMap.
 - The page explores different attempts determining what works and doesn't while exploring the why and bench-marking existing hashmaps and comparing them to the final result.
 - Achieves significant improvements over `std::unordered_map` being nearly 3x faster
- hofministryga.org (<https://www.hofministryga.org>)
 - The webpage for The Household of Faith Ministry in Georgia.
 - It uses SQLite to maintain recorded messages and serve them to visitors.
- CMM (<https://github.com/hofministryga/CMM>)
 - A database management system of The Household of Faith Ministry in Georgia to add, edit, and remove recorded messages without having to use the command line.
 - Written in C++, unit tested using Catch2, and uses an older version of SMPL (GLIB) for its GUI and may no longer compile.
 - Auto updates on run.
- IHex (<https://github.com/Davidah121/IHex>)
 - The foundation for a future interpreted language that is intended to run sand boxed alongside C / C++ code.
 - Lightweight and simple to understand and include in projects only needing a few files while achieving close to Java speeds before its JIT compiler kicks in and being faster than Lua and Python.
- SEMML (<https://github.com/Davidah121/SEMML>)
 - SIMD Extended Math Library. It adds in useful functions that are cross platform for all other x86 based CPUs that support SSE but are not Intel based products.
- SimpleSerialization (<https://github.com/Davidah121/SimpleSerialization>)
 - A simple to use Serializer for C++ classes and structures. Adds basic reflection as well.
- CppBench2 (<https://github.com/Davidah121/CppBench2>)
 - A low cost profiler for C++ applications that requires very little adjustments to source code and can easily be disabled.